

Title

QuartoManual: A Starter Template and Framework for Notebook-Driven Scientific Software Operations Manuals

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Abstract

Climate-Smart Public Health (CSPH) is a data-driven approach to detect, quantify, and adapt to climate-related health impacts at a global scale, spanning surveillance, risk assessment, early warning and response, and healthcare systems adaptation (Golden et al., 2025). In practice, CSPH depends on computational workflows that integrate heterogeneous climate, environmental, and health data and must remain understandable, reproducible, and adaptable across collaborators and local settings.

Despite the central role of such workflows in academia and industry, effective, reproducible software documentation remains overlooked (Aghajani et al., 2019). This challenge is likely to intensify as software development by both humans and AI continues to expand in scale and complexity (Martinelli, 2026; Robbes et al., 2026). Developers therefore need solutions beyond static documentation to support workflow execution, verification, and reproducibility in practical, user-friendly ways.

Conventional documentation can describe a workflow, but for complex projects it is often unable to capture and verify runtime details such as platform dependencies, configuration, and stepwise relationships (Ebert et al., 2015). Command-line interface (CLI) wrappers can improve automation and reproducibility by providing a direct interface for complex tasks (van der Aalst et al., 2009), but they introduce a persistent trade-off: specialized wrappers improve consistency but can be rigid, while generic wrappers shift complexity to the user through numerous flags and configuration options (Brack et al., 2022; Sadiq et al., 2005). AI and agentic systems are also promising, but their reproducibility remains limited by non-determinism, and current workflow implementations still require careful human oversight to achieve acceptable reliability (Siddiq et al., 2025; Agrawal et al., 2026).

*Executable literate programming approaches can encode complex workflow steps while generating idempotent results (Knuth, 1992). Quarto is a modern, plain-text, multi-language, multi-output framework for creating reproducible documents and workflows (Allaire et al., 2026). Building on Quarto, we introduce **Quarto Manuals: a framework for executable software manuals that guide users through workflows as ordered, interactive pages combining explanation, code execution, and verification**. Rather than hiding workflow decisions, Quarto Manuals keep them visible and editable, making them well-suited to research computing contexts where local constraints and expert judgment shape execution. We demonstrate the approach with three examples of increasing sophistication: staging datasets; geospatial aggregation of environmental exposure data; and containerizing geospatial data science environments on HPC clusters. Across these cases, Quarto Manuals aim to improve the reproducibility and reliability of scientific workflows, particularly in contexts like CSPH where local constraints and expert judgment shape execution.*

For authors, they provide a structured yet adaptable way to turn recurring workflows into reusable operators' manuals; for operators, they provide a stepwise, testable process for generating reproducible setup and execution artifacts and confirming progress in a structured, auditable way. In future work, we will explore how Quarto Manuals can complement AI and

agentic systems by helping keep computational workflows transparent, verifiable, and reproducible as they grow in complexity.

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Connection to Mission, Goals, & Interests of US-RSE Community

Scientific programming workflows are becoming more complex, while existing support mechanisms often remain poorly matched to that complexity. The data is only expected to grow in volume, variety, and velocity, and a lingering question in the face of AI is, “are we really going to stop writing code for this?”

We see Quarto Manuals as a practical contribution to this problem and directly aligned with the research interests of the US-RSE community. Research software engineers are often responsible not only for building software, but also for making workflows understandable, reproducible, maintainable, and usable by others. Quarto Manuals are designed to support that work by combining explanation, code execution, and verification in an ordered, operator-facing format.

This contribution has three main highlights. First, Quarto Manuals contributes a reusable technical resource for documenting and delivering complex scientific workflows. Second, it supports the visibility and value of research software engineering by treating workflow design, execution, and handoff as first-class engineering artifacts rather than incidental burdens or purely local practice. Third, it serves the broader community by making it easier for collaborators with different levels of programming experience to run, inspect, and adapt computational workflows in a structured and auditable way.

Perhaps most importantly, Quarto Manuals are also intended to be complementary — not adversarial — to the growing use of AI-assisted programming. As AI-generated code becomes more common, research teams will increasingly need workflow structures that make human-written instructions and LLM-generated code easier to inspect, test, reproduce, and maintain together. In that sense, Quarto Manuals may provide a useful harness for incorporating AI-assisted programming into scientific workflows, without sacrificing rigor or transparency. Ultimately, we hope Quarto Manuals helps make scientific programming more accessible, reproducible, and inclusive, while supporting the high standards of software engineering and collaboration that the US-RSE community promotes.

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